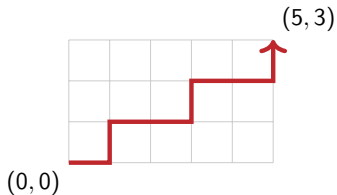


Some Essential Problems

Section 2.1

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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Today's lecture

1. Sum and Product Rules

2. Permutations and Selections

3. Counting Subsets

4. Poker Hands

What exactly are we counting?

Question

How many ways can we choose three objects from a collection of ten?

The answer depends on the rules:

- Does order matter?
- Can an object be used more than once?
- Are the objects distinguishable?

The sum rule

Sum rule

If a set of outcomes is partitioned into disjoint classes of sizes $a_1, \dots, a_r$, then its size is $a_1 + \dots + a_r$.

Example

A committee chooses either one of 8 first-year students or one of 6 second-year students. There are $8 + 6 = 14$ choices.

Why disjointness matters

Question

There are 20 students taking algebra and 15 taking combinatorics. How many take at least one of the courses?

Adding $20 + 15$ counts a student in both courses twice.

The sum rule applies directly only when the classes are disjoint. We will later correct overlaps using inclusion and exclusion.

The product rule

Product rule

Suppose a procedure has r successive stages, with a_i choices at stage i for every possible set of earlier choices. Then there are

$$a_1 a_2 \cdots a_r$$

complete outcomes.

The actual choices may depend on earlier stages; their number must be the stated a_i .

A product-rule example

Example

How many three-digit positive integers have distinct digits?

The first digit has 9 choices. The second has 9 choices, including zero but excluding the first digit. The last has 8 choices.

$$9 \cdot 9 \cdot 8 = 648.$$

Permutations

Permutation

A permutation of n distinct objects is an ordering of all of them.

Counting permutations

There are

$$n! = n(n-1) \cdots 2 \cdot 1$$

permutations. We use $0! = 1$.

Ordered selections without repetition

Falling factorial

For $0 \leq k \leq n$,

$$n^{\underline{k}} = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

This counts ordered selections of k distinct objects from n objects. In particular, $n^{\underline{0}} = 1$.

Ordered selections with repetition

Counting words

The number of words of length k over an alphabet of n symbols is n^k .

Proof

Each of the k positions has n choices. Apply the product rule.

Example

There are 2^k binary words of length k and 10^k strings of k decimal digits.

Unordered selections

Binomial coefficient

The number of k -element subsets of an n -element set is denoted by $\binom{n}{k}$.

Formula

For $0 \leq k \leq n$,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Why divide by $k!$?

Proof

There are n^k ordered selections of k distinct objects. Each k -element subset appears once for every ordering of its elements, hence exactly $k!$ times. Therefore

$$\binom{n}{k} = \frac{n^k}{k!}.$$

Division works because every desired outcome has the same number of ordered representations.

Committees and officers

Question

From 12 people, choose a committee of 4, and then choose a chair and secretary from that committee.

First the committee, then the officers:

$$\binom{12}{4} \cdot 4 \cdot 3.$$

First the officers, then the other members:

$$12 \cdot 11 \binom{10}{2}.$$

A first double count

$$\binom{12}{4} \cdot 4 \cdot 3 = 12 \cdot 11 \binom{10}{2}.$$

Both sides count exactly the same objects in different orders.

This method often proves identities more naturally than factorial manipulation.

Counting all subsets

Theorem

An n -element set has 2^n subsets.

Proof

For each element, independently decide whether to include it. There are two choices for each of n elements.

Grouping subsets according to size also gives

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

Selections with repetition

Question

How many ways are there to choose k objects from n types when repetition is allowed and order does not matter?

An outcome is specified by nonnegative integers

$$x_1 + \cdots + x_n = k.$$

We will derive $\binom{n+k-1}{k}$ using stars and bars and generating functions.

Four basic models

	No repetition	Repetition allowed
Ordered	$\frac{n!}{(n-k)!}$	n^k
Unordered	$\binom{n}{k}$	$\binom{n+k-1}{k}$

Here the n available objects or types are distinguishable.

Circular arrangements

Theorem

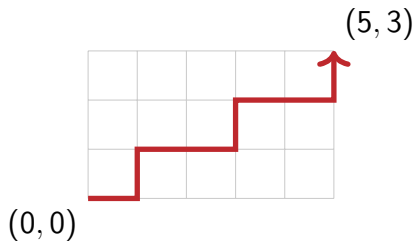
There are $(n - 1)!$ circular arrangements of n distinct people, when rotations are identified and reflections are different.

Proof

Fix one specified person in a reference seat. Arrange the remaining $n - 1$ people around the circle.

Changing the equivalence rule changes the counting problem.

Lattice paths



Question

How many paths from $(0, 0)$ to $(5, 3)$ use only right and up steps?

Encoding a path

Proof

Every path has 5 right steps and 3 up steps. Choose the 3 positions occupied by up steps among the 8 steps:

$$\binom{8}{3} = 56.$$

In general, the number of such paths to (a, b) is $\binom{a+b}{a}$.

Counting by complements

Example

How many binary words of length 10 contain at least one 1?

There are 2^{10} words in total. Exactly one, the all-zero word, has no 1.

$$2^{10} - 1 = 1023.$$

A difficult positive condition can have a simple complement.

Poker hands

The model

A standard deck has 52 distinct cards: 13 ranks and 4 suits. A poker hand is an unordered set of 5 cards.

The total number of hands is

$$\binom{52}{5} = 2,598,960.$$

Four of a kind

Example

Choose a hand with four cards of one rank and a fifth card of a different rank.

Choose the repeated rank in 13 ways. All four suits are forced. Choose the fifth card from the 48 cards of other ranks.

$$13 \cdot 48 = 624.$$

Full houses

Example

A full house has three cards of one rank and two of another rank.

Choose the triple rank, its suits, the pair rank, and its suits:

$$13 \binom{4}{3} \cdot 12 \binom{4}{2} = 3744.$$

The two ranks have different roles, so they are chosen in order.

Exactly one pair

Choose the pair rank and its two suits. Choose three distinct other ranks, and one suit for each:

$$13 \binom{4}{2} \binom{12}{3} 4^3 = 1,098,240.$$

The requirement that the other ranks be distinct excludes two pairs, three of a kind, and full houses.

Choosing three arbitrary cards after choosing the pair would overcount and include unwanted hands.

Flushes

Example

A hand all of whose cards have the same suit can be chosen in

$$4 \binom{13}{5} = 5148$$

ways.

This includes straight flushes. Under the usual poker classification, exclude the 10 consecutive rank patterns in each of 4 suits:

$$5148 - 40 = 5108.$$

Question

How many 5-element subsets of $\{1, \dots, 12\}$ contain exactly two elements of $\{1, 2, 3, 4\}$?

Identify two independent choices before calculating.

Proof

Choose 2 of the first 4 elements and 3 of the remaining 8:

$$\binom{4}{2} \binom{8}{3} = 6 \cdot 56 = 336.$$

The two choices determine the subset uniquely.

- Use the sum rule for disjoint alternatives and the product rule for successive choices.
- Specify whether order and repetition matter.
- Divide only when every outcome has the same multiplicity.
- Encode objects as words, subsets, or paths whenever possible.

Binomial Coefficients